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RESEARCH ARTICLE

Detecting and Identifying Insider Threats Based on Advanced Clustering Methods

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ABSTRACT This paper explores the analysis of user behavior in information systems through audit records, creating a behavior model represented as a graph. The model captures actions over a specified period, facilitating real-time comparison to identify insider threats exploring anomalies detected in behavior models. "e-StepControl," developed by "ABC software" Ltd., incorporates this approach for monitoring user behavior in different business environments. The study proposes enhancing this solution with automatic user clustering, achieved by grouping individuals exhibiting similar behavior patterns using AI/ML algorithms. The research evaluates various clustering methods, discussing their suitability for grouping users based on their behavior. The subsequent step involves leveraging user class behavior models to identify anomalies by comparing an individual's actions with the behavior model expected in their specific user group. This extension aims to enhance the system's ability to detect potentially malicious activities, providing data security administrators with timely alerts in case of deviations from typical behavior.

INDEX TERMS Anomaly detection, clustering algorithms, data mining, information system user behavior analysis, information technology security, insider threats detection.

I. INTRODUCTION

The examination of an individual's activities within an IT information system (IS) involves an analysis of the actions logged in the system's audit records. This analysis results in the creation of a behavioral model representing the typical behavior of the user while using specific IS-s, presented as a graph. By accumulating behavioral data over a specific timeframe, it becomes possible to compare recent user actions to their established behavioral model. This process helps to determine the consistency of their current behavior with their usual behavior patterns, enabling the detection of irregularities or deviations in their actions. In cases where security breaches are possible, these anomalies trigger immediate alerts to the data security administrator.

Authors have already implemented the detection of such anomalies in the behavior of IS users in the e-StepControl solution and introduced it for monitoring of user behavior in various business environments [1], [2], [3]. The idea of this research paper is to expand the existing solution with clustering of users, which essentially involves automatically grouping of IS users with identical or similar behavioral models into separate groups (classes) using Artificial Intelligence and Machine Learning (AI/ML) algorithms. The behavioral model of corresponding user group can then be used to identify anomalies by comparing the individual IS user behavior against the user group behavioral model.

Everyone's work within an IS can be analyzed by examining the audit logs of user actions conducted within the IS. This analysis involves constructing a behavioral model specific to each individual, represented in the form of a graph, tailored to the business functionality of each particular

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IS. The behavioral model associated with each user group can then be aimed at identifying anomalies by comparing an individual user behavior against the corresponding group model.

Overall, this extension of the e-StepControl solution will assist with identifying cases where, for example, a user account data have been stolen without their knowledge, and someone else is operating within the IS using the user identity. It will also help to identify situations where the users themselves may be violating their authorized system usage due to certain circumstances. There are cases when IS users are granted extensive access to accumulated data, but this access should only be utilized for fulfilling work responsibilities [4].

In such cases, by comparing the current user behavior model with the typical behavior model of respective user group, the IT security administrator will receive information about potentially malicious activities of a particular IS user and can investigate the potentially malicious behavior incidents.

The proposed algorithm for identifying incidents in the research does not address cases involving the base authorization mechanism. If there is a legitimate need to grant a user extensive access to an IS, there is a risk that the user may engage in unsanctioned activities within the boundaries of their authorization, thereby violating their privileges while operating the IS.

Therefore, the expansion of the e-StepControl solution, as proposed in this research paper, is relevant for any IS (in a business environment) that contains personal, business-critical, sensitive, or restricted-access data. It serves as a tool to help IT security administrators in monitoring the usage of data stored in the IS and be promptly informed of any potentially malicious activities [5].

Until now, grouping of users, based on similar behavior, has been a manual process which is time-consuming and does not always yield satisfactory results. Well-based grouping of users is crucial for achieving high precision in anomaly detection [6].

II. RESEARCH PROBLEM AND RELATED WORK

The development of IS-s and their expanding usage possibilities, which allow for solving various business problems and supporting operations in different industries, brings an increasing challenge – the security threats of unauthorized data usage that can come from internal system users and be equivalent to those posed by external threats (hackers, etc.) [7], [8].

The traditional technique to control the access of ‘‘sanctioned’’ IS users to data is to grant them access rights and control the utilization of these rights within the framework of authorization. However, such access control, or authorization mechanisms, only partially address the tasks of controlling information accessibility and usage. Therefore, alternative solutions need to be implemented to mitigate the risks of unauthorized data usage [9].

To mitigate security risks, the monitoring of IS usage based on analyzing the behavioral models of users can be utilized. The company ‘‘ABC software’’ Ltd. has implemented this approach in the product e-StepControl [1], which is already applied for different problem domains and not only for IS users threats identification [10]. The conceptual scheme of the algorithm used in e-StepControl solution for identification of potentially malicious sessions with Markov chains is shown in Fig. 1 [11].

While analyzing the behavior of a specific user within a particular IS [12], a user behavior model is created, which includes all the actions performed by this user in the IS during a certain period, as well as the probability of execution of sequential actions pairs, which is calculated based on the principles defined by Markov chains [13]. Furthermore, each new sequence of actions performed by the user (user session) is analyzed in comparison with this behavior model. The presence of sequentially executed actions pairs in each analyzed session affects the anomaly coefficient of that session, considering the probability values of such pairs in the user behavior model [14]. If the anomaly coefficient of the session exceeds a predefined threshold, that session is identified as potentially malicious, and is handed over to a security expert for detailed analysis.

Hence, the necessity arises to classify users into proper categories. At present, this classification process relies on manual efforts. To improve efficiency and make better use of time and human resources, it becomes imperative to automate this procedure using formal algorithms [10].

One more essential difference of the approach offered in the paper is in the clustering object itself. Two main directions in the approaches using K-means algorithm for IS internal threats (user behavior anomalies) identification try to identify abnormal IS users' behavior by clustering behavior actions and looking for anomalies in action clusters. They first extract a set of statistical features from user behavior data, such as the distribution of login times, file access patterns, and system calls. Then these features are used to train a clustering model to represent these usage patterns. Finally, the model analyzes the clusters to identify anomalous clusters of users' behavior that may indicate insider threats. These approaches for clustering models use either actions, or sequences of events. In contrast, we offer to cluster users into groups with similar behavior, based on the similar or the same set of actions. And then we are looking for anomalies of users behavior inside each cluster separately.

The common method of categorization of IS users requires manual parsing and analysis of extensive lists containing users roles. This involves the search for resemblances in sets of positions or responsibilities and subsequently grouping users who share identical roles. Nonetheless, this manual approach can give rise to several challenges, including:

1) Time-consuming process – manual user grouping becomes notably time-consuming, especially when dealing with large IS-s containing a high volume of

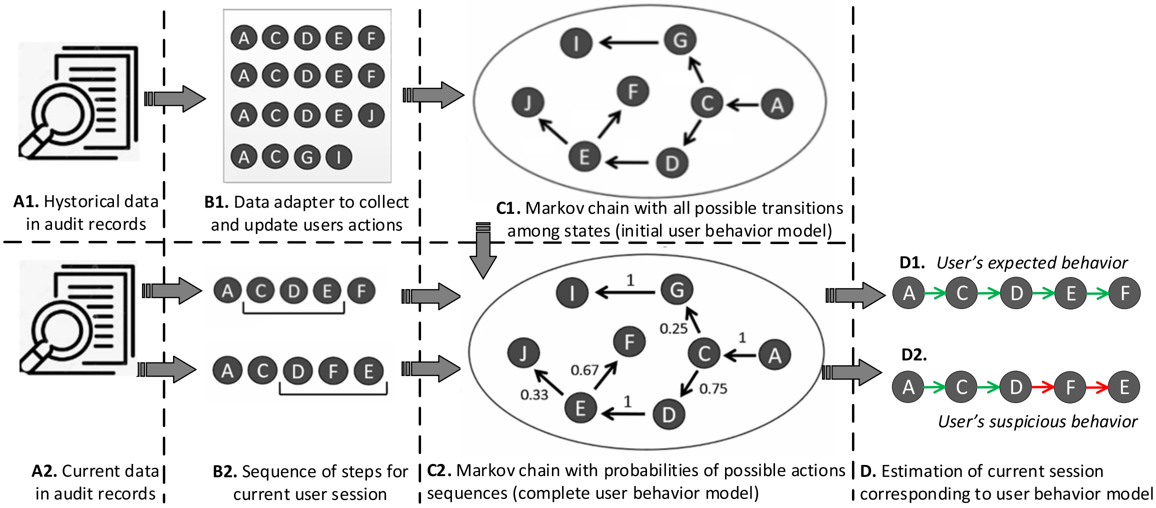

FIGURE 1. The overall workflow of the data in e-StepControl solution.

users. The need for multiple iterations further increases the time commitment required.

2) Incompatibility with complex IS-s – the traditional manual approach, which relies on studying of user roles lists, may prove inadequate for complex and sizable IS-s. This can result in improper grouping and introduce security vulnerabilities.
3) Precision limitations – relying solely on roles list analysis for grouping may lead to errors, as multiple users may be linked to different roles and perform diverse actions, posing security risks. Consequently, security administrators may struggle to accurately define compositions of users groups.
4) Overly large user groups – manual methods can yield excessively large users groups, uniting users from multiple existing roles or groups. This further complicates anomaly analysis tasks
5) Overlooked users with combined roles – users with combined roles may be occasionally overlooked during manual grouping. Human analysts may encounter challenges in efficiently assessing and making objective decisions about grouping these combined roles.
6) Oversight of users with excessive permissions – manual approaches may fail to identify users who have been erroneously included in groups, particularly when they have been assigned overly broad permissions beyond their job responsibilities.

As a result, there is a pressing need to utilize AI/ML and clustering mechanisms to address these challenges and achieve more precise and efficient grouping of IS users. Employing AI/ML methodologies allows for the automated recognition and categorization of users based on their behavioral and habitual similarities. Furthermore, these approaches facilitate the creation of groups of users with similar characteristics, prevent the formation of overly broad users groups, and aid in identifying instances of excessive authorization granted to specific users.

In the section that follows, existing clustering methods and algorithms are analyzed to select the most suitable algorithm for solving the task of IS users grouping / clustering.

III. CHOOSING A METHOD AND ALGORITHM FOR GROUPING IS USERS

The clustering task, also known as cluster analysis, is a statistical classification technique where a set of objects or data points with similar properties is grouped into clusters. It involves various methods and algorithms used to group similar objects into their respective categories. The goal of cluster analysis is to organize observed data into logical structures to extract meaningful information [15].

The adaptation of existing clustering approaches specifically for grouping IS users, based on their behavior within multiple IS-s, is complex, because it requires to cluster not just abstract sets of objects or data points, but rather the users themselves, who perform multiple actions within IS-s, considering the sequence of their actions. Therefore, in this research paper, an "embedded" (two-level attribute) clustering approach will be applied that combines both the set of user actions and the set containing information about transitions between these actions.

The types of clustering methods commonly used in AI/ML are referenced in [16] and [17] and described below. The advantages and disadvantages of each clustering method in the context of grouping IS users with the same or similar behavior are listed in Table 1.

Partitioning clustering, which is also referred to as centroid-based clustering, is a data clustering technique that partitions data into non-hierarchical groups. It is one of the most often used clustering methods for data grouping. The Partitioning clustering starts by defining the initial number of clusters and iteratively assigns input data points to clusters based on their distances from the cluster centers (typically using the Euclidean distance). In each iteration, a specific data point is assigned to the closest cluster in terms

TABLE 1. Advantages and disadvantages of clustering methods in the context of grouping of IS users with the same or similar behavior.

Clustering method	Advantages of the method	Disadvantages of the method	Representatives
Partitioning clustering	The method can work with high-dimensional feature vectors as the principles of distance calculation do not significantly change based on the number of dimensions used.	In cases when the desired number of clusters in the dataset is unknown beforehand, the result may not be optimal, as the required number of resulting clusters needs to be specified before starting the algorithm.	K-means, K-medoids (PAM - Partitioning around medoids)
	The method's algorithms provide clear and interpretable results. The created clusters are well defined, easy to understand and analyze.	Partitioning-based algorithms assume that each data point belongs to exactly one cluster, making them unsuitable for scenarios with overlapping classes of objects.	
	The method can discover isolated and rare clusters (assuming the input data object feature vectors do not contain errors), which can be particularly useful in various business domains.	The method's algorithms can be sensitive to noise and erroneous data. Noise points can disrupt the clustering process and negatively affect the quality of the result (e.g., creating a separate cluster with a single erroneous object, etc.).	
Hierarchical clustering	Hierarchical clustering can be used with various distance metrics, allowing for customization of algorithms based on specific data properties.	High computational complexity, especially when analyzing large datasets. The time and memory resources required for clustering can be significant.	Agglomerative Hierarchical Clustering, Divisive Hierarchical Clustering, Single-linkage clustering, Complete-linkage clustering
	The method is relatively robust against noise and outliers. Values that significantly differ from other values are usually isolated in their own branches of the dendrogram.	The method assumes a hierarchical data structure, which may not always reflect reality. It can be problematic for data with complex objects relationships or perceptions, resulting in suboptimal clustering results for such data.	
	Hierarchical clustering can handle different types of data, such as numerical, categorical, or mixed-type data.	The method produces a dendrogram, which can be difficult to interpret when objects have multiple attributes.	
Density-based clustering	Density-based clustering algorithms can handle clusters with varying densities, identifying both "dense" and "sparse" regions in the data and assigning points accordingly.	Lack of "global perspective" – density-based clustering algorithms mainly focus on local density and neighborhood relationships. As a result, there may be difficulties in identifying the global structure of the datasets, leading to suboptimal clustering results.	DBSCAN (Density-Based Spatial Clustering of Applications with Noise), OPTICS (Ordering Points to Identify the Clustering Structure), DENCLUE (DENsity-based CLUstEring)
	Density-based clustering algorithms provide intuitive boundaries for resulting clusters based on the density of data points. Therefore, these results are relatively easy to interpret.	Density-based clustering algorithms may struggle with processing datasets containing overlapping clusters. If the density of two or more clusters is similar or if there is a significant overlap in their regions, the algorithm may have difficulties separating these accurately.	
	Density-based clustering algorithms preserve spatial information of the data considering distances between data points, which can be useful in applications where spatial relationships between data points are important.	Difficulties in processing high-dimensional data – in high-dimensional spaces, the concept of density becomes less meaningful, which can diminish the usefulness of clustering results.	
Distribution model-based clustering	These algorithms provide flexibility in choosing different distribution models to represent clusters, allowing for more precise handling of various complex data patterns.	The respective algorithms assume that the data follow a specific statistical distribution (such as Gaussian distribution, etc.). If the data do not comply with these assumptions, the clustering results may be inaccurate.	Gaussian Mixture Models (GMM), Expectation-Maximization (EM) Algorithm, Bayesian information criterion
	Distribution model-based clustering algorithms have the ability to handle missing data, i.e., datasets with incomplete or partial observations.	Clustering algorithms of this method can suffer from overfitting issues, which can lead to situations where the clustering result fails	

TABLE 1. *(Continued.)* **Advantages and disadvantages of clustering methods in the context of grouping of IS users with the same or similar behavior.**

		to correctly identify the underlying patterns or structure in the data.	
	Distribution model-based clustering can provide a measure of uncertainty in cluster assignment for each data point.	These algorithms can be computationally intensive, especially for large datasets or complex distribution models.	
Fuzzy clustering	This method provides flexible and overlapping membership of data points in multiple clusters. This can be particularly useful when working with complex datasets that may not have clear cluster boundaries.	The algorithms often require specifying various parameters, such as fuzziness coefficient, convergence criteria, distance metrics, etc. The choice of these parameters can be subjective and significantly affect the clustering results.	Fuzzy C-means (FCM) algorithm, Fuzzy subtractive clustering, Gustafson-Kessel Algorithm
	Such clustering is well suited for handling data uncertainty and ambiguity. It is suitable for situations where there is a degree of uncertainty or imprecision in the data.	The method is not suitable for situations where the unambiguous membership of each input data object to specific clusters needs to be determined.	
	The method is robust to various data transformations, such as data normalization or dimensionality reduction of object attributes, making it suitable for a wide range of data preprocessing techniques.	Compared to other existing classical clustering algorithms, fuzzy clustering is based on relatively less explored mathematical principles.	
Grid-based clustering	Grid-based clustering algorithms are relatively easy to implement, making these accessible to a wide range of users.	Grid-based clustering algorithms can be sensitive to the choice of grid size. Choosing an inappropriate grid size can result in inaccurate clustering results.	STING, WaveCluster, CLIQUE
	These algorithms are inherently easy to parallelize, allowing for efficient use of available computational resources.	The algorithms are designed to determine clusters with approximately rectangular shapes. There may be difficulties in accurately identifying clusters with complex shapes.	
	The method can efficiently handle incremental updates to the dataset since examining neighboring cells and updating existing clusters is not a complex process.	The randomly chosen starting point of the grid coordinate system can negatively affect clustering results.	
Graph-based clustering	This method can discover complex relationships between data points, including nonlinear relationships or dependencies.	Graph-based clustering may not be as efficient or applicable to data that lack a network/topological structure.	Highly Connected Subgraphs (HCS), Maximal Clique Enumeration, CosTaL
	Graph-based clustering can be used to uncover "communities" in networks, which can be useful in social network analysis, identification of functional modules in biological networks, etc.	Most clustering algorithms assume that the input data are complete. Incomplete data can lead to subjective or inaccurate clustering results.	
	The results of clustering can be visualized using graph layouts and drawing libraries, providing a visually interpretable representation of objects relationships.	Due to the requirements of graph-based clustering methods in terms of computational resources and memory footprint, difficulties may arise when processing large datasets with complex topologies.	

of distance. Then, the cluster centers are updated based on the newly assigned data points. The partitioning clustering continues until a defined condition is met, such as when data points no longer change clusters during the iterations or when the maximum specified number of iterations is reached. As a result, each cluster center is created to minimize the distance between the data points within the same cluster, compared to the points of other cluster centers [18].

Hierarchical clustering is an approach employed to group similar data points. It creates a hierarchical cluster structure by iteratively merging or splitting clusters based on similarity metrics. There are two main approaches to hierarchical clustering: agglomerative and divisive clustering. Agglomerative clustering starts with each data point as a separate cluster and merges them based on their similarity, while divisive clustering starts with all data points in one cluster and recursively divides them into smaller clusters. The resulting hierarchical structure can be visualized using a dendrogram, which shows the merging or splitting of clusters [19].

Density-based clustering is a technique that clusters data points according to their density within the object space. The main principle of density-based clustering is to identify dense regions of data points separated by regions of lower density. Within the method, core points are identified, which are data points with a sufficient number of neighboring points within a specified distance threshold. These core points form initial clusters, contents of which are gradually expanded and refined during the execution of corresponding algorithm [20].

Distribution model-based clustering is a data analysis method employed to detect clusters by relying on the underlying statistical probability distribution models, such as Gaussian distribution or Poisson distribution, among others. The goal of the method is to estimate the parameters of the distribution models for each cluster. These calculations are typically performed using Bayes' theorem. Once the distribution models are estimated, the data is partitioned based on the probability that objects in the dataset belong to a specific distribution [21].

Fuzzy clustering, also known as fuzzy cluster analysis, represents a form of soft clustering technique wherein data points can simultaneously have membership in multiple clusters, which distinguishes it from various other clustering algorithms. The main principle of fuzzy clustering methods is to assign a membership value to each data point, reflecting the degree to which the point belongs to each cluster. These values are typically represented as probabilities or degrees of membership ranging from 0 to 1. The process involves iteratively updating the membership values and cluster centers until convergence is achieved. The updating of membership values is based on the distances between data points and cluster centers, with closer points are assigned higher membership values [22].

Grid-based clustering is a clustering method that organizes data points into groups based on their proximity within a spatial grid. This method involves dividing the data space into grid cells, with each cell representing a small rectangular region of the data space. The first step is to determine the size of the grid cells. Once this is done, each input data point is assigned to the corresponding cell based on its spatial coordinates. Then, the clustering algorithm examines the content of each cell and its neighboring cells to create clusters of objects [23], [24].

Graph-based clustering is a method employed to group similar data points by considering connectivity of these within a graph structure. The main principle is that data points (vertices) that are tightly connected in the graph are likely to belong to the same cluster. The connections between points (graph edges) reflect similarity or distance between these. The graph is constructed based on the similarity or distance matrices of input data objects. Once the graph is built, clustering is performed by dividing the graph into subgraphs. This is usually achieved by making cuts that increase the similarity within clusters and reduce the similarity between these [25].

Upon a comprehensive examination of the various clustering methods and a thorough evaluation of corresponding approach, suitability, advantages, and limitations, several key conclusions have been drawn, especially in the context of the specific task of grouping users on the basis of their behavior within the IS. Corresponding methods and reasons for potential limitations of these are as follows:

- Fuzzy clustering is not a suitable choice for the dataset in this study due to the necessity to unambiguously identify IS user membership within specific group of users. Implementing fuzzy clustering may result in outcomes that are less interpretable. In this scenario users may exhibit partial affiliations with multiple clusters, rendering it impossible to derive clear insights into user behavior patterns. Consequently, this approach denies the selection of a particular behavioral model for validating user sessions.
- Distribution model-based clustering is not a proper choice because the input data solely comprise actual user actions and their sequences, making it challenging to define the requisite probability distribution. Selecting an appropriate distribution model for IS user behavior is difficult due to the complex modeling assumptions associated with distribution model-based methods, such as assuming if user behavior corresponds to specific probability distributions (e.g., Gaussian, Bayesian, etc.). These assumptions can be difficult to validate, potentially failing to accurately capture underlying behavioral patterns within a particular IS. Incorrect model selection results in non-optimal clustering outcomes. Furthermore, distribution model-based methods often demand data preprocessing, including feature scaling and dimensionality reduction, to fulfill the probabilistic distribution assumptions.
- Graph-based clustering is not a feasible option because there are no clearly defined topological relationships among the research objects to be clustered. Graph-based clustering abstracts away the sequential nature of the data by treating transitions between actions as undirected edges, disregarding the temporal sequence of actions. This abstraction may lead to the loss of valuable information related to action sequences, a critical aspect in comprehending IS user behavior.
- Hierarchical clustering is not a suitable approach as there is no necessity to identify hierarchical data structures in the input data. Hierarchical clustering generates dendrogram structures resembling trees, wherein clusters can nest within one another. This can complicate the interpretation of hierarchy and the selection of a specific behavioral model for comparison with a group of users. Additionally, hierarchical clustering tends to be less scalable, especially when dealing with many data points, such as sequences of actions of users within complex IS-s. The hierarchical algorithm typically exhibits higher time complexity, rendering it less efficient for large datasets.

On the other hand, the clustering methods that hold potential for our research task (as these accept attribute vectors describing the objects as input data), are density-based clustering, grid-based clustering and partitioning clustering. Further analysis of these three clustering methods has led to the following conclusions:

- Density-based clustering presents challenges in determining suitable density and distance parameters for our specific context. For instance, algorithms like DBSCAN require setting density and distance thresholds, which can be complex to define precisely in the realm of user behavior data in IS-s. Additionally, density-based algorithms often assume that cluster objects exhibit similar densities, which may not hold true for specific user behavior patterns in IS-s.
- Grid-based clustering necessitates predefining the size of grid cells as one of its parameters, a task made complicated by the variability in actual behavior of users within the IS and the granularity of these actions.Other limitations, such as the arbitrary choice of the grid coordinate system's starting point and better compatibility with sets having approximately rectangular spatial forms, further restrict the usefulness of this method.
- Partitioning clustering has such advantages, as speed, support for large attribute vectors, the ability to identify specific clusters with unique objects, and ease of result interpretation; however, it also has a drawback in the classical algorithm version – the need to specify the number of resulting clusters [26]. This presents a challenge since the number of clusters is unknown in the context of grouping of IS users. However, a potential solution to this issue has been identified with corresponding theoretical justification (see Section IV below).

Based on these analysis results, the decision has been made to employ the Partitioning clustering method for our research task.

IV. DETECTING AND IDENTIFYING INSIDER THREATS ALGORITHM

Partitioning clustering methods represent a widely adopted set of algorithms in the field of data mining. Their primary objective is to divide the initial dataset into K clusters, with the aim of grouping similar data points while maximizing the distinctions between these clusters [27], [28]. Among the classical algorithms in partitioning clustering, two noteworthy approaches are K-means [29] and K-medoids, also known as "PAM – Partition Around Medoids" [30]. The key distinction between these methods lies in the choice of cluster centers: K-means employs geometric centroids, while K-medoids uses actual data points within the cluster, which may or may not coincide with the geometric mean. After careful consideration of both approaches, the decision was made to specifically employ the K-means algorithm due to the following reasons:

- Outlier sensitivity – K-means exhibits lower sensitivity to outliers compared to K-medoids. Outlying points have a greater impact on K-medoids, as these directly influence the selection of the cluster center point. In contrast, K-means relies on geometric centroids (based on the average value of all points in the cluster), rendering it more robust against potential outliers.
- Computational efficiency – K-means demonstrates higher computational efficiency, especially when dealing with large datasets. Calculating the geometric center requires fewer computational resources than choosing a specific point from multiple candidate points within a cluster.
- Convergence – the K-means algorithm guarantees convergence to (at least local) optimum when sufficient number of iterations are performed, enhancing its reliability in finding a stable clustering solution.

A. K-MEANS ALGORITHM FOR GROUPING USERS

The sub-steps of the K-means algorithm are as follows:

1) Specify the desired number K of resulting clusters.
2) Randomly initialize the centroids of these clusters.
3) Assign each data point to its nearest centroid.
4) Recalculate the centroid coordinates by computing the average distortion value assigned to each cluster.
5) Repeat steps 3 and 4 until the assignment of points clusters no longer changes or until the maximum allowed number of iterations is reached.
6) Return the resulting K clusters and the points assigned to these.

However, a significant challenge with the K-means algorithm is the need to determine the number of resulting clusters in the initial step of the algorithm [31]. This is problematic when automating the grouping of IS users based on their behavior, as the exact number of groups is unknown in advance. Within the method's implementation, this issue is addressed by utilizing the "Elbow method" [32].

The "Elbow method" involves empirically identifying the best partitioning of objects (in this case – IS users) by iteratively evaluating the "inertia" values for different numbers of clusters, aggregating the results, and selecting the optimal outcome.

The "Elbow method" predicts that the most significant information gain (and, consequently, the most meaningful division of objects into classes) occurs when the rate of change in the "inertia" values undergoes the fastest shift.

Therefore, the K-means algorithm refined with "Elbow method" is selected for clustering in the task of grouping of IS users based on their behavior defined by actions performed within IS and their sequential pairs. Overall schematic representation of algorithm steps is shown in Fig. 2.

Input data (1) on user-performed sessions in the particular IS are provided for the creation of a matrix of user activities and pairs of these (2), as well as for the establishment of an individual user behavior model (5a) using the Markov chain generation algorithm.

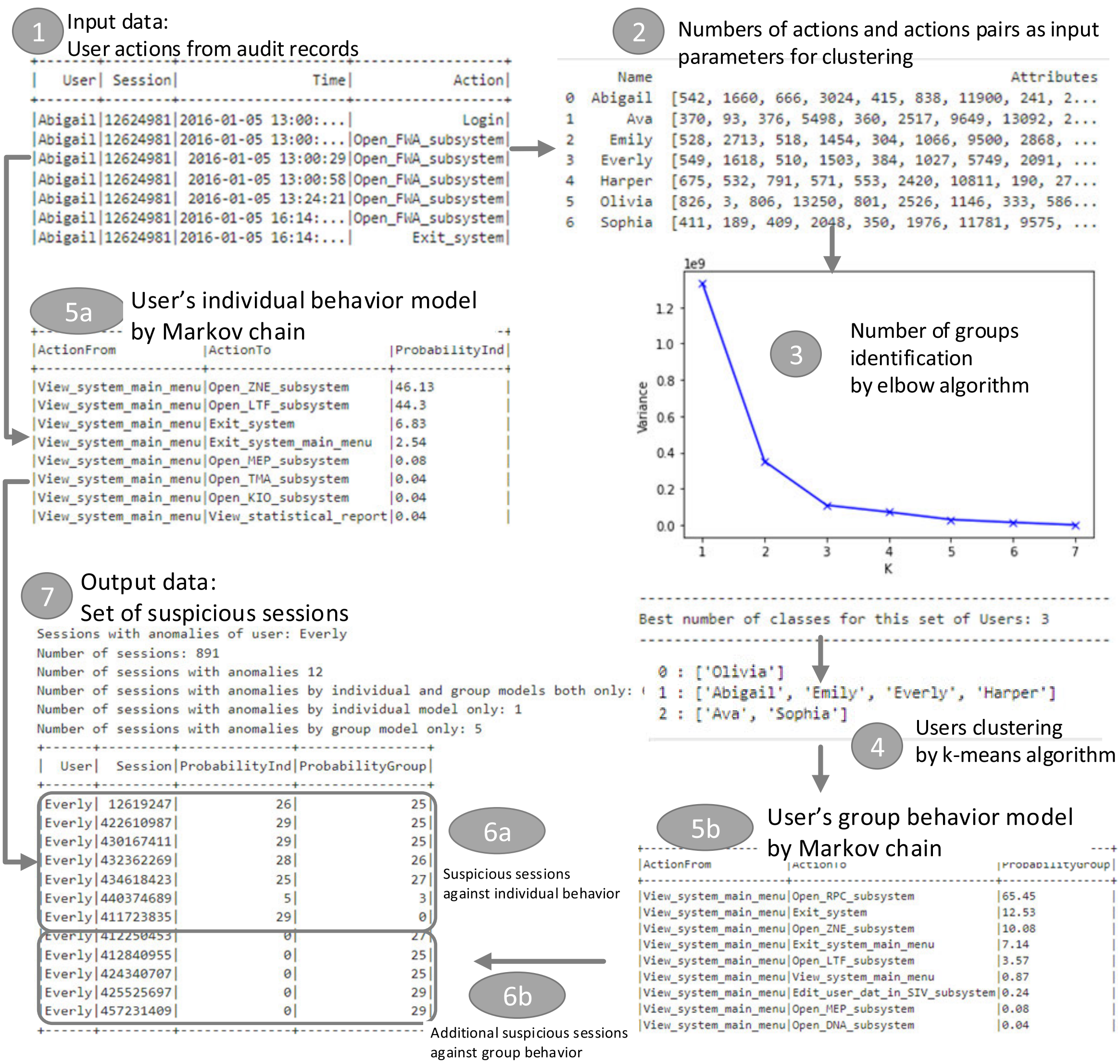

FIGURE 2. Schematic representation of algorithm steps.

The matrix of behavior parameters is utilized to divide users into groups (4) using the K-means algorithm, with the number of groups determined by the "Elbow method" (3). For each group, a user behavior model specific to that group (5b) is created using the Markov chain generation algorithm.

B. USERS BEHAVIOR ANALYSIS

The behavioral model for individual users and each group encompasses all unique actions carried out by IS users within that group, including the transitions between these actions and the associated transition probabilities, which are computed using the principles of a Markov chain [13].

If the expectation factor of a session goes beyond a predetermined threshold, it raises a red flag as potentially malicious. This suggests that the session's behavior deviates from what is typically observed in the majority of other sessions within the same user group.

Consequently, we can identify sessions that are flagged as potentially malicious by both the individual behavior model and the group behavior model (designated as "7" in Fig. 2). We can also identify sessions flagged as potentially malicious by only one of the models: either the individual behavior model (labeled as "6a" in Fig. 2) or the group behavior model (referred to as "6b" in Fig. 2). In the schematic representation in Fug. 2, steps labeled as "5a" and "6a" pertain to the analysis against the individual behavior model, while steps labeled as "5b" and "6b" correspond to the analysis against the group behavior model.

To validate the algorithm's functionality, real data have been used, which contain actions performed by employees of different roles and multiple branches/departments engaged in data entry, review, customer communication, report generation, etc. From the initial data, information about 488 users who have performed approximately 2.8 million (2,776,770) actions, was selected (while ensuring

Group	UserName	UserTrust%	SessionCount	1	2	3	4
0	Eleanor	38.34	193	6	0	8	0
0	Ella	43.75	240	11	1	15	1
0	Elizabeth	44.7	302	11	2	16	2
0	Amelia	45.71	466	24	1	51	1
0	Luna	46.18	262	11	3	22	3
1	Olivia	50.81	803	5	0	5	0
3	Mia	51.84	463	5	0	13	0
3	Sophia	56.89	399	7	0	11	0
3	Ava	71.43	364	7	0	6	0
0	Evelyn	71.77	496	9	10	47	9
0	Camila	73.28	625	15	7	33	21
0	Scarlett	74.01	558	3	14	11	85
0	Emily	76.48	523	11	1	4	15
0	Isabella	79.72	779	12	19	23	18
0	Harper	81.3	631	13	7	34	4
0	Everly	84.02	513	5	1	4	5
2	Emma	85.34	771	20	4	35	11
0	Sofia	95.67	554	10	0	6	3
2	Charlotte	98.15	1135	16	11	9	8
0	Abigail	98.4	562	5	10	9	9

FIGURE 3. **The number of potentially malicious sessions for all users relative to the total number of sessions.**

```
-RECORD 0-----------------------------
 GroupId                        | 2
 UserId                         | 18
 UserName                       | Ava
 UserTrustPercent               | 71.43
 SessionCount                   | 364
 IndividualExtremeSessionCount  | 7
 IndividualAverageSessionCount  | 0
 GroupExtremeSessionCount       | 6
 GroupAverageSessionCount       | 0

All sessions of user Ava ( 364 ):
Suspicious sessions of user Ava ( 9 ):
```

SessionId	SessionTrust%	StepCount	SuspiciousStepCount	Duration(sec)
421604017	21.0	64	8	20648
425336101	24.0	188	13	31063
419185045	24.0	150	13	30237
418901337	24.0	232	2	30060
448419987	27.0	194	1	31518
461634621	33.0	150	1	11327
415026009	49.0	102	1	6477
446248013	27.0	157	4	25342
456583471	27.0	63	6	26246

FIGURE 4. **Suspicious sessions of the user "Ava."**

```
Dictionary:
1 : Login
2 : Initialize_tasks_list
3 : View_system_main_menu
4 : Open_ZNE_subsystem
5 : Open_LTF_subsystem
6 : Open_SEP_subsystem
7 : Invite_new_user_to_a_chat
8 : Send_email
9 : Exit_system
```

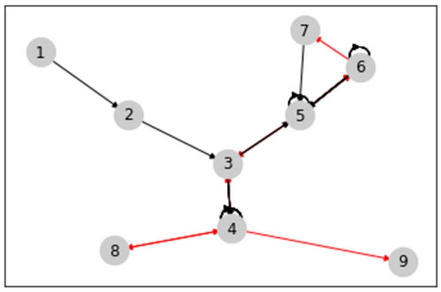

FIGURE 5. **A graph with the most suspicious session with the ID "421604017" for the user "Ava."**

overall data integrity). For this paper, names of users were changed to random female names, real actions names were changed to traditional actions performed in convenient IS-s (like "generate_report", "login/exit", "open/close", "view main_menu", etc.).

In the experiment involving seven users, analyzing their real actions against the expected group behavior, additional potentially malicious sessions were identified for three users. In the experiment with 20 users, analyzing their real actions, additional potentially malicious sessions were identified for eight users. In the experiment with 488 users, grouping the users from the previous experiments, the same groups were maintained, and potentially malicious sessions were identified for 158 users only after analyzing the group behavior model.

For the above experiments, the number of potentially malicious sessions for all users relative to the total number of sessions for each user is shown in Fig. 3. As can be seen from Fig. 3, in some cases the number of potentially malicious sessions relative to the group model is greater than the same number relative to the individual one. Numbered columns in the table in Fig. 3 show the corresponding numbers of sessions identified by the particular algorithm. 1, 3 denote numbers of suspicious sessions identified on the basis of user individual behavior taking general and extreme values for expectation factor, while 2, 4 represent numbers of suspicious sessions identified by user group behavior also taking general and extreme values for expectation factor.

Criteria, corresponding coefficients of suspiciousness for each criterion and a set of sessions of user "Ava" are demonstrated in Fig. 4.

For this user "Ava", a graph was outlined for the most suspicious session with the ID "421604017" (with the same coefficient, and fewer steps) – see Fig. 5.

On the graph in Fig. 5, steps 4-9 are potentially suspicious, but acceptable in general (although, it is relatively rare for a user to open the main menu of the IS and exit immediately). However, steps 6-7 and 4-8 are potentially malicious and unacceptable for this particular problem area, which indicates that a security incident has been identified.

C. REAL-WORLD IMPLICATIONS

While existing approaches have made strides in mitigating insider threats, there remains a need for innovative methods that can adapt to evolving attack vectors and organizational dynamics. K-means algorithm application refined with elbow method as it is introduced in this paper, leverages identification of insider threats highlighting potentially malicious sessions with suspicious steps, based on Markov chain behavioral model created for users clusters. Building upon previous research in anomaly detection and clustering algorithms, our solution detects suspicious behavior among authorized users.

The results of our study have significant implications for real-world scenarios in various business environments. By accurately identifying anomalous behavior among authorized users, the algorithm offered in the paper and integrated into e-StapControl solution empowers organizations to proactively mitigate insider threats before they escalate into security breaches. In high-stakes industries such as finance, healthcare, and government, where the consequences of insider attacks can be dire, the ability to detect and respond to suspicious behavior in real-time is paramount. Moreover, by integrating e-StapControl within existing security frameworks and users log data, organizations can enhance their overall cybersecurity posture and bolster regulatory compliance efforts.

While our solution shows promise in insider threat detection, several implementation challenges must be addressed to realize its full potential across different business environments. Firstly, the diversity of organizational structures, data sources, and security policies necessitates a flexible and customizable approach to implementation. Adapting the solution to accommodate varying data formats, access controls, and user behaviors requires careful consideration and collaboration between security practitioners and IT stakeholders. Additionally, the scalability and resource requirements of the solution may pose challenges for organizations with limited computational resources or legacy systems. Balancing the need for robust threat detection with operational constraints and resource limitations remains a key challenge in implementing our solution effectively. By understanding the implications of our results and addressing implementation challenges, organizations can harness the full potential of e-StepControl to safeguard their critical assets and mitigate insider threats effectively.

V. CONCLUSION

Clustering algorithms are commonly employed to partition data into distinct clusters within various problem domains. In the realm of anomaly detection within information systems' user behavior, these algorithms serve to group data points based on their similarities, thereby isolating anomalous behavior. Existing research endeavors encompass a diverse array of datasets (ranging from log files to network traffic and malware data) each necessitating tailored adaptations of clustering algorithms for effective anomaly detection. All data sets are unique, and the clustering algorithms must be adapted to certain data set to be effective. Each algorithm has its own strengths and weaknesses, and the choice of algorithm will depend on the specific data set and the desired outcome of the anomaly detection process.

The algorithm offered in this paper also varies. Authors propose to take log data consisting of users' performed actions and to create users' clusters based on similar actions set and further to look for employees suspicious actions as less performed in comparison with the employees in the same cluster. Within the scope of the study, the developed method is planned to be used in the existing solution "e-StepControl" for both to create new functionality and replace manual system initialization tasks with AI/ML-based automated algorithms.

Importantly, this method is poised to be integrated into the existing solution "e-StepControl." Beyond merely augmenting functionality, the proposed algorithm aims to streamline system initialization tasks by replacing manual processes with AI/ML-driven automation. By doing so, it not only enhances detection capabilities but also improves operational efficiency within the context of insider threat management.

As a result of the study, the following conclusions were drawn:

1) Grouping of users works reliably: experiments have been conducted to groups of 7, 20 and 488 users. As the number of users increases, the users from previous experiments are placed in the same groups, indicating correct distribution of users. Knowing the source of the data, it can be assumed that a larger group of users is more likely to consist of users that work in a IS in a similar manner.
2) There is a particular user who is always placed in a separate group (even from all $\sim$500 users), and this user is always the only one in that resulting group. This suggests that this user might be a technical specialist or an administrator or a manager. The fact of constant presence of such user in an according group indicates that the algorithm is working correctly.
3) Grouping of users is useful, because analysis of sessions based on the group behavior model identifies additional potentially malicious sessions that are not detected while analyzing the individual user behavior model.
4) It was not possible to determine in the available data how many of the identified different behavior sessions were real security incidents. However, the fact that users perform something atypical not only for an individual user but for the entire group of users, gives reasons to suspect that unauthorized potentially malicious actions were really performed.

Future work on how IS user clustering can help in identification of potentially malicious behavior is concerned with experiments aimed at analysing user behavior against behavior model of the group of users defined for this particular user, but without his/her sessions. This can help in situations when the user regularly performs non-allowed actions, and thus, pollutes the group behavior model with such activities.

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the development of innovation clusters'' implementation rules within the competence centers'' framework.

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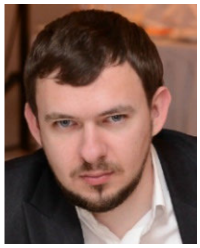

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